



**COMPETENCY STANDARD**

**FOR**

**ADVANCED PATTERN MAKING WITH CAD**

**(RMG SECTOR)**

**Skills for Industry Competitiveness and Innovation Program (SICIP)**  
**Finance Division, Ministry of Finance**



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The Competency Standards for Advanced Pattern Making with CAD (RMG sector) is a document for the development of curricula, teaching and learning materials, and assessment tools. It also serves as the document for providing training consistent with the requirements of the industry for individuals who pass through the set standard via assessment would be qualified and settled for a relevant job.

The document was developed under the Skills for Employment Investment Program (SEIP) and subsequently reviewed and updated with the engagement of occupation-specific industry experts to use in training under the **Skills for Industry Competitiveness and Innovation Program (SICIP)**. This document is owned by the Finance Division of the Ministry of Finance of the People's Republic of Bangladesh.

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## INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

**The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.**

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

## OVERVIEW:

A **Competency Standard** is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Competency Standard has been developed by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

## Units & Elements at a Glance:

### Generic Competencies (30 Hrs.)

| Code               | Unit of Competency                                                   | Elements of Competency                                                                                                                                                                                                                                                        | Duration (Hours) |
|--------------------|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| SICIP-RMG-APM-01-G | Apply occupational health and safety (OHS) practice in the workplace | <ol style="list-style-type: none"><li>1 Identify OHS policies and procedures.</li><li>2 Apply personal health and safety practices.</li><li>3 Report hazards and risks.</li><li>4 Respond to emergencies</li></ol>                                                            | 08               |
| SICIP-RMG-APM-02-G | Manage personal and professional development                         | <ol style="list-style-type: none"><li>1 Interpret personal development skills.</li><li>2 Set and meet self-development priorities.</li><li>3 Maintain professional growth and development.</li></ol>                                                                          | 10               |
| SICIP-RMG-APM-03-G | Support innovation and manage change                                 | <ol style="list-style-type: none"><li>1 Identify needs for innovation in the workplace</li><li>2 Apply creative approach and solution</li><li>3 Support flexible and innovative ways of working</li><li>4 Adapt to emerging technological changes and opportunities</li></ol> | 12               |
| <b>Total Hour</b>  |                                                                      |                                                                                                                                                                                                                                                                               | <b>30</b>        |

**Sector Specific Competencies (20 hrs.)**

| <b>Code</b>        | <b>Unit of Competency</b>           | <b>Elements of Competency</b>                                                                                                                                                 | <b>Duration (Hours)</b> |
|--------------------|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| SICIP-RMG-APM-01-S | Apply Quality Procedures            | <ol style="list-style-type: none"><li>1. Identify quality procedures</li><li>2. Follow quality procedure</li><li>3. Maintain standard procedure</li></ol>                     | 10                      |
| SICIP-RMG-APM-02-S | Apply green practices in RMG sector | <ol style="list-style-type: none"><li>1. Interpret green concepts</li><li>2. Minimize resource use in the workplace</li><li>3. Implement waste management practices</li></ol> | 10                      |
| <b>Total Hour</b>  |                                     |                                                                                                                                                                               | <b>20</b>               |

## Occupation Specific Competencies (222 hrs.)

| Code               | Unit of Competency                          | Elements of Competency                                                                                                                                                                                                                                                                              | Duration (hours) |
|--------------------|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| SICIP-RMG-APM-01-O | Interpret advanced pattern making using CAD | <ol style="list-style-type: none"> <li>1. Understand advanced pattern making.</li> <li>2. Identify advanced garment items.</li> <li>3. Identify critical parts, points and measuring guide of garments.</li> </ol>                                                                                  | 20               |
| SICIP-RMG-APM-02-O | Perform pattern making with CAD             | <ol style="list-style-type: none"> <li>1. Apply basic sewing techniques</li> <li>2. Perform pattern making for menswear.</li> <li>3. Perform pattern making for children wear.</li> <li>4. Perform pattern making for lingerie wear.</li> <li>5. Maintain quality issues of the pattern.</li> </ol> | 152              |
| SICIP-RMG-APM-03-O | Perform pattern grading                     | <ol style="list-style-type: none"> <li>1. Interpret size and measurement differences.</li> <li>2. Apply rule table for grading.</li> <li>3. Carry out pattern grading.</li> </ol>                                                                                                                   | 20               |
| SICIP-RMG-APM-04-O | Make marker consumption and printing        | <ol style="list-style-type: none"> <li>1. Make marker.</li> <li>2. Calculate consumption.</li> <li>3. Print and plot marker using CAD system.</li> </ol>                                                                                                                                            | 30               |
| <b>Total Hours</b> |                                             |                                                                                                                                                                                                                                                                                                     | <b>222</b>       |



## COMPETENCY STANDARD: ADVANCED PATTERN MAKING WITH CAD

### Generic Competencies

|                                                                                                                                                                                                                                                                                                                                                                  |                                     |                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------|
| <b>Unit of Competency:</b><br><b>APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICE IN THE WORKPLACE</b>                                                                                                                                                                                                                                                        | <b>Nominal Duration:</b><br>08 hrs. | <b>Unit Code:</b><br>SICIP-RMG-APM-01-G |
| <b>Unit Descriptor:</b><br><br>This unit covers the skills, knowledge and attitudes required to apply occupational health and safety (OHS) practices in the workplace. It specifically includes the tasks of identifying OHS policies and procedures, applying personal health and safety practices, reporting hazards and risks, and responding to emergencies. |                                     |                                         |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                        | Performance Criteria                                                                                                                                                                                                                       |
|-----------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Identify OHS policies and procedures       | 1.1 <b><u>OHS policies</u></b> and safe operating procedures are interpreted<br>1.2 Safety signs and symbols are identified and followed.<br>1.3 Response, evacuation procedures and other contingency measures are interpreted correctly. |
| 2. Apply personal health and safety practices | 2.1 OHS policies and procedures are interpreted in the workplace including <b><u>personal protective equipment (PPE)</u></b> .<br>2.2 Common health issues are recognized.<br>2.3 Common safety issues are identified.                     |
| 3. Report hazards and risks                   | 3.1 Hazards and risks are identified.<br>3.2 Hazards and risks assessment and controls are interpreted                                                                                                                                     |
| 4. Respond to emergencies                     | 4.1 Respond to alarms and warning devices.<br>4.2 <b><u>Emergency response plans and procedures</u></b> are responded to.<br>4.3 <b><u>First aid procedures</u></b> during emergency situations are identified.                            |

### Range of variables:

| Variables                                  | Range (may include but not limited to)                                                                         |
|--------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| 1. OHS policies                            | 1.1 Organizational OHS policies<br>1.2 International OHS requirements<br>1.3 Fire safety rules and regulations |
| 2. Emergency response plans and procedures | 1.1 Firefighting procedures<br>1.2 Earthquake response procedures                                              |

|                                  |                                                                                                                              |
|----------------------------------|------------------------------------------------------------------------------------------------------------------------------|
|                                  | 1.3 Emergency response plans and procedures<br>1.4 Medical and first aid                                                     |
| 3. First aid procedure           | 1.1 Washing of open wound<br>1.2 Washing chemically infected area<br>1.3 Applying bandage<br>1.4 Taking appropriate medicine |
| 4. Personal protective equipment | 1.1 Safety glasses<br>1.2 Ear plugs<br>1.3 Gloves<br>1.4 Apron<br>1.5 Helmet<br>1.6 Mask<br>1.7 Safety shoes                 |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                               |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning knowledge | 1.1 Workplace OHS policies and procedures<br>1.2 Work safety procedures<br>1.3 Emergency response procedures:<br>1.3.1 Firefighting<br>1.3.2 Earthquake response<br>1.3.3 Accident response<br>1.4 Types of hazards (biological, chemical and physical) and their effects<br>1.5 OHS awareness<br>1.6 Personal protective equipment (PPE)<br>1.7 Malfunctions and resolutions |
| 2. Underpinning skill     | 2.1 Identifying OHS policies and procedures<br>2.2 Applying personal health and safety practices<br>2.3 Reporting hazards and risks<br>2.4 Responding to emergencies                                                                                                                                                                                                          |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincere and honest to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Communication with peers, subordinates and seniors in the workplace           |
| 4 Resource implications   | The following resources must be provided:<br><br>3.1 Workplace (simulated or actual)<br>3.2 Personal protective equipment (PPE)<br>3.3 Firefighting equipment<br>3.4 Emergency response manual<br>3.5 First aid kits<br>3.6 Stationery                                                                                                                                        |

|  |                      |
|--|----------------------|
|  | 3.7 Learning manual. |
|--|----------------------|

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical aspects of competency | <p>Assessment required evidence that the candidate:</p> <ul style="list-style-type: none"> <li>1.1 identified OHS policies and procedures</li> <li>1.2 applied personal health and safety practices</li> <li>1.3 reported hazards and risks</li> <li>1.4 responded to emergencies.</li> </ul>                         |
| 2. Methods of assessment          | <p>Competency should be assessed by:</p> <ul style="list-style-type: none"> <li>2.1 Written test</li> <li>2.2 Practical Demonstration</li> <li>2.3 Oral Questioning</li> <li>2.4 Portfolio (Optional)</li> </ul>                                                                                                      |
| 3. Context of assessment          | <ul style="list-style-type: none"> <li>3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.</li> <li>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</li> </ul> |

|                                                                                                                                                                                                                                                                                                                                    |                                     |                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------|
| <b>Unit of Competency:</b><br><b>MANAGE PERSONAL AND PROFESSIONAL DEVELOPMENT</b>                                                                                                                                                                                                                                                  | <b>Nominal Duration:</b><br>10 hrs. | <b>Unit Code:</b><br>SICIP-RMG-APM-02-G |
| <b>Unit Descriptor:</b><br><br>This unit covers the skills, knowledge and attitudes required to manage personal and professional development. It specifically includes the tasks of interpreting personal development skills, setting and meeting self-development priorities and maintaining professional growth and development. |                                     |                                         |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                          | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Interpret personal development skills        | 1.1 Objectives of personal development skills are described.<br>1.2 <b><u>Personal development skills</u></b> are identified.<br>1.3 Intra and Interpersonal relationships are maintained in the course of managing oneself.<br>1.4 Self-analysis is performed and personal development needs are identified.                                                                                                                              |
| 2. Set and meet self-development priorities     | 1.5 Tasks are prioritized to achieve personal, team and organizational goals and objectives.<br>1.6 <b><u>Resources</u></b> are utilized efficiently and effectively to manage work priorities and commitments.<br>1.7 Economic usage and maintenance of facilities are followed as per established procedures.                                                                                                                            |
| 3. Maintain professional growth and development | 1.8 Pro activeness/zeal is demonstrated in fulfilling personal and professional growth requirements.<br>1.9 <b><u>Trainings and career opportunities</u></b> are identified and accessed based on job requirements.<br>1.10 <b><u>Recognitions</u></b> are sought/ received and demonstrated as proof of career advancement.<br>1.11 <b><u>Licenses and/or certifications</u></b> relevant to the job and career are obtained and renewed. |

### Range of variables:

| Variables                      | Range (may include but not limited to)                                                                                  |
|--------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 1. Personal Development skills | 1.1 Problem-solving<br>1.2 Self-confidence<br>1.3 Adaptability<br>1.4 Integrity<br>1.5 Work ethic<br>1.6 Pro-activeness |
| 2. Resources                   | 1.7 Human                                                                                                               |

|                                       |                                                                                                                                                                                                    |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                       | 1.8 Financial<br>1.9 Technology                                                                                                                                                                    |
| 3. Trainings and career opportunities | 1.10 Participation in training programs<br>3.1.1 Technical<br>3.1.2 Supervisory<br>3.1.3 Managerial<br>3.1.4 Continuing Education<br>1.11 Serving as Resource Persons in conferences and workshops |
| 4. Recognitions                       | 1.12 Recommendations<br>1.13 Citations<br>1.14 Certificate of Appreciations<br>1.15 Commendations<br>1.16 Awards<br>1.17 Tangible and Intangible Rewards                                           |
| 5. Licenses and/or certifications     | 1.18 National Certificates<br>1.19 Certificate of Competency<br>1.20 Support Level Licenses<br>1.21 Professional Licenses                                                                          |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning knowledge | 1.1 Importance of personal development skills.<br>1.2 Organizational policies relevant to training and professional growth.<br>1.3 Company operations, procedures and standards.<br>1.4 Resources in work environment.                                                                                                                                              |
| 2. Underpinning skill     | 1.5 Utilizing and improving personal development skills<br>1.6 Maintaining Intra and Interpersonal relationship<br>1.7 Utilizing communication skills<br>1.8 Prioritizing tasks in accordance with work commitment.<br>1.9 Utilizing resources efficiently and effectively                                                                                          |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincere and honest to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Communication with peers, subordinates and seniors in the workplace |
| 4. Resource implications  | The following resources must be provided:<br><br>1.1 Workplace (simulated or actual)<br>1.2 Workplace procedures<br>1.3 Standard operating procedure<br>1.4 Workplace documents, signs and symbols<br>1.5 Codes of conduct                                                                                                                                          |

|  |                     |
|--|---------------------|
|  | 1.6 Projector       |
|  | 1.7 Stationery      |
|  | 1.8 Learning manual |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical aspects of competency | <p>Assessment required evidence that the candidate:</p> <ul style="list-style-type: none"> <li>1.1 attained personal development skills.</li> <li>1.2 maintained intra and interpersonal relationship in the course of managing oneself.</li> <li>1.3 prioritized tasks according to work commitments.</li> <li>1.4 identified training and career opportunities.</li> <li>1.5 completed trainings based on the requirements of the industries.</li> <li>1.6 acquired and maintained licenses and/or certifications according to the requirement of the qualification.</li> </ul> |
| 2. Methods of assessment          | <p>Competency should be assessed by:</p> <ul style="list-style-type: none"> <li>1.7 Written test</li> <li>1.8 Practical Demonstration</li> <li>1.9 Oral Questioning</li> <li>1.10 Portfolio (Optional)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                 |
| 3. Context of assessment          | <ul style="list-style-type: none"> <li>1.11 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.</li> <li>1.12 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</li> </ul>                                                                                                                                                                                                                                                           |

|                                                                                                                                                                                                                                                                                                                                                                                |                                     |                                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------|
| <b>Unit of Competency:</b><br><b>SUPPORT INNOVATION AND MANAGE CHANGE</b>                                                                                                                                                                                                                                                                                                      | <b>Nominal Duration:</b><br>12 hrs. | <b>Unit Code:</b><br>SICIP-RMG-APM-03-G |
| <b>Unit Descriptor:</b><br><br>This unit covers the skills, knowledge and attitudes required to support innovation and manage change. It specifically includes the tasks of identifying needs for innovation in the workplace, applying creative approach and solution, supporting flexible and innovative ways of working and adapting to emerging changes and opportunities. |                                     |                                         |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                                       | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Identify needs for innovation in the workplace            | 1.1. The need for <b><u>innovation</u></b> in scope of work is recognized<br>1.2. The value of <b><u>innovative practices</u></b> in the workplace is recognized<br>1.3. Existing way of working in the workplace are interpreted.<br>1.4. Drawbacks of existing way of working are identified.<br>1.5. Benefits of change are identified to make it consistent with organizational requirements<br>1.6. Realistic timelines and targets for implementation of changes are set. |
| 2. Apply creative approach and solution                      | 2.1 Opportunities and creative approaches to implement innovation practices are identified<br>2.2 Creative approaches of coworkers pertaining to work practices are analyzed and incorporated<br>2.3 Innovation in accordance with organizational requirements changes are implemented                                                                                                                                                                                          |
| 3. Support flexible and innovative ways of working           | 3.1 Maximum innovative opportunities are promoted.<br>3.2 Work assignments to facilitate innovative work skills are organized.<br>3.3 Team members are provided with guidance and coaching on innovation in the workplace.<br>3.4 Models of innovative work practice are provided and discussed<br>3.5 Appropriate environment for learning and innovation is maintained.                                                                                                       |
| 4. Adapt to emerging technological changes and opportunities | 4.1 Usages of different <b><u>technologies</u></b> is determined based on job requirements.<br>4.2 Appropriate technology is selected as per work specifications.<br>4.3 Relevant technology is effectively used in carrying out functions.                                                                                                                                                                                                                                     |

|  |                                                                        |
|--|------------------------------------------------------------------------|
|  | 4.4 Appropriate implementation tools are used as per task requirement. |
|--|------------------------------------------------------------------------|

#### Range of variables:

| Variables               | Range (may include but not limited to)                                                                                                                                               |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Innovative practices | 1.1. Self-directed support to implement new ideas<br>1.2. Collaborative arrangement to apply new method of working<br>1.3. Making scope of work more efficient to use new technology |
| 2. Innovation           | 1.4. New ideas<br>1.5. Different ideas<br>1.6. New methods of doing work<br>1.7. Use of new Technology                                                                               |
| 3. Technologies         | 1.8. Office technology<br>1.9. Industrial technology<br>1.10. System technology<br>1.11. Information technology<br>1.12. Training technology                                         |

#### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning knowledge | 1.1. Definition of innovation<br>1.2. Current practice in own scope of work<br>1.3. Workplace procedures<br>1.4. Support required to generate creative ideas<br>1.5. Difference between innovation and creativity<br>1.6. Common effects of change and innovation in the workplace<br>1.7. Industrial and organizational context of change<br>1.8. Organization's policies, plans, procedures and structure<br>1.9. Knowledge of resources required by the organization's operations<br>1.10. Awareness on technology and its function<br>1.11. Operating instructions<br>1.12. Applicable software<br>1.13. Company policy in relation to relevant technology<br>1.14. Technology adaptability |
| 2. Underpinning skill     | 1.1. Identifying resources required for creativity and innovation<br>1.2. Contributing to brainstorming session<br>1.3. Identifying issues and concerns of one's scope of work<br>1.4. Encouraging co-workers to generate and develop ideas<br>1.5. Evaluating potential obstacles to and opportunities for                                                                                                                                                                                                                                                                                                                                                                                     |



|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | creativity and innovation<br>1.6. Sharing of best practices related to innovation and creativity<br>1.7. Applying relevant technology<br>1.8. Using update machine and Software<br>1.9. Acquiring troubleshooting skills Identifying issues<br>1.10. Identifying current industry standard diagnostic tools<br>1.11. Describing common malfunctions and resolutions.<br>1.12. Determining the root cause of a routine malfunction |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincere and honest to duties<br>3.4 Eagerness to learn<br>3.5 Tidiness and timeliness<br>3.6 Environmental concerns<br>3.7 Respect for rights of peers and seniors at workplace<br>3.8 Communication with peers and seniors at workplace                                                                                     |
| 4. Resource implications  | The following resources must be provided:<br>4.1 Workplace (simulated or actual)<br>4.2 Projector<br>4.3 Stationery<br>4.4 Learning manual                                                                                                                                                                                                                                                                                        |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                   |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical aspects of competency | Assessment required evidence that the candidate:<br>1.1 identified need for innovation in the workplace<br>1.2 applied creative approach and solution<br>1.3 supported flexible and innovative ways of working<br>1.4 adapted to emerging technological changes and opportunities |
| 2. Methods of assessment          | Competency should be assessed by:<br>2.1 Written test<br>2.2 Practical Demonstration<br>2.3 Oral Questioning<br>2.4 Portfolio (Optional)                                                                                                                                          |
| 3. Context of assessment          | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.                     |

## The Sector Specific Competencies

|                                                                                                                                                                                                                                                                 |                                     |                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------|
| <b>Unit of Competency:</b><br><b>APPLY QUALITY PROCEDURES</b>                                                                                                                                                                                                   | <b>Nominal Duration:</b><br>10 hrs. | <b>Unit Code:</b><br>SICIP-RMG-APM-01-S |
| <b>Unit Descriptor:</b><br><br>This unit covers the skills, knowledge and attitudes required to apply quality procedures. It specifically includes the tasks of identifying quality procedures, following quality procedure and maintaining standard procedure. |                                     |                                         |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency         | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Identify quality procedures | 1.1. <b><u>Manuals</u></b> and guidelines for RMG production processes are collected as per sample specifications.<br>1.2. The importance of manuals in ensuring quality standards is recognized.<br>1.3. Instructions and procedures for quality control are identified.<br>1.4. Required information is collected from manuals.<br>1.5. Performance measurement systems are identified.                                                                           |
| 2. Follow quality procedure    | 2.1 Instructions and procedures are followed strictly and duties are performed in accordance with demand of <b><u>quality improvement system</u></b> .<br>2.2 Concept of supplying product or service to meet the <b><u>customer quality requirements</u></b> is understood and accordingly applied.<br>2.3 Conformance to specifications is ensured.<br>2.4 Defects are detected and reported to the relevant authority according to standard operating procedures |
| 3. Maintain standard procedure | 3.1 Performance is assessed at regular interval.<br>3.2 Specifications and standard operating procedures are established<br>3.3 Quality of product is checked and verified.<br>3.4 Quality control and quality assurance system procedures for each job are followed.<br>3.5 Conformance to specification is ensured in every case at all situations.                                                                                                               |

### Range of Variables

| Variable   | Range                                                                                         |
|------------|-----------------------------------------------------------------------------------------------|
|            | May include but not limited to:                                                               |
| 1. Manuals | 1.1 Buyers' specification manual<br>1.2 Compliance manual<br>1.3 Maintenance procedure manual |

|                                  |                                                                                                                             |
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|                                  | 1.4 Periodic maintenance manual<br>1.5 Quality manual<br>1.6 Signs and symbols, instruction manuals                         |
| 2. quality improvement system.   | 2.1. Quality inspection<br>2.2. Testing<br>2.3. Quality control.<br>2.4. Quality assurance<br>2.5. Total Quality Management |
| 3. customer quality requirements | 3.1. Performance<br>3.2. Features<br>3.3. Reliability<br>3.4. Conformance<br>3.5. Aesthetics<br>3.6. Durability             |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Importance of maintaining quality<br>1.2 quality, quality assurance, quality control, quality inspection, quality improvement and total quality control.<br>1.3 Process and procedures for improving and maintaining quality<br>1.4 Procedures for addressing defects.<br>1.5 Record keeping within the quality improvement system in workplace<br>1.6 Factors, which affect successful implementation of the quality systems and procedures. |
| 2. Underpinning Skills    | 2.1. Maintaining good quality<br>2.2. Eliminating poor quality<br>2.3. Understanding the meaning of the key terms - quality, quality assurance, quality control, quality inspection, quality improvement and total quality control.<br>2.4. Improving and maintaining quality<br>2.5. Addressing defects and procedures<br>2.6. Recording within the quality improvement system in workplace.<br>2.7. Implementing quality systems and procedures |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincere and honest to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in workplace<br>3.8 Communication with peers, sub-ordinates and seniors in workplace                                                                                      |
| 4. Resource Implications  | 4.1 Workplace (simulated or actual)                                                                                                                                                                                                                                                                                                                                                                                                               |

|  |                                                                          |
|--|--------------------------------------------------------------------------|
|  | 4.2 Map/globe<br>4.3 Projector<br>4.4 Stationery<br>4.5 Learning manual. |
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### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                               |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 identified quality procedures<br>1.2 followed quality procedure<br>1.3 maintained standard procedure                                                                                                  |
| 2. Methods of Assessment          | Competency should be assessed by:<br>2.1 Written test<br>2.2 Practical Demonstration<br>2.3 Oral Questioning<br>2.4 Portfolio (Optional)                                                                                                                      |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert. |

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|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------|
| <b>Unit of Competency:</b><br><b>APPLY GREEN PRACTICES IN RMG SECTOR</b>                                                                                                                                                                                                                    | <b>Nominal Duration:</b><br>10 hrs. | <b>Unit Code:</b><br>SICIP-RMG-APM-02-S |
| <b>Unit Descriptor:</b><br><br>This unit covers the skills, knowledge and attitudes required for apply green practices in RMG sector. It specifically includes the tasks of interpreting green concepts, minimizing resources in the workplace and implementing waste management practices. |                                     |                                         |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                     | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Interpret green concepts                | 1.1 <b><u>Principles of green practices in RMG (Ready-Made Garment)</u></b> manufacturing are explained.<br>1.2 Key terms and symbols in RMG production processes are identified.<br>1.3 <b><u>Sources of environmental impacts</u></b> during RMG manufacturing activities are identified.<br>1.4 RMG manufacturing activities use are explained.<br>1.5 <b><u>Ways of mitigating environmental impacts</u></b> in RMG manufacturing are explained. |
| 2. Minimize resources use in the workplace | 2.1 Water, energy, and raw material consumption in RMG manufacturing are documented.<br>2.2 <b><u>Recyclable and non-recyclable items</u></b> in RMG production processes are identified.<br>2.3 Procedures to reduce resource consumption in RMG manufacturing are implemented.<br>2.4 Sustainable alternatives in RMG manufacturing are explored and applied.                                                                                      |
| 3. Implement waste management practices    | 3.1 <b><u>Different types of waste in RMG manufacturing</u></b> are identified.<br>3.2 Hazardous waste in RMG production is disposed of according to environmental regulations.<br>3.3 Green habits to reduce waste in personal and professional life are practiced.                                                                                                                                                                                 |

### Range of Variables

| Variable                                                     | Range                                                                                                                                                                                                                             |
|--------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                              | May include but not limited to:                                                                                                                                                                                                   |
| 1. Principles of green practices in RMG (Ready-Made Garment) | 1.1 Reducing energy consumption in RMG manufacturing processes.<br>1.2 6R for waste management in RMG production:<br>1.2.1 Refuse unnecessary materials and processes.<br>1.2.2 Reduce resource consumption and waste generation. |

|                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|--------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                  | <p>1.2.3 Reuse materials within the production process.</p> <p>1.2.4 Recycle materials for future use in RMG manufacturing.</p> <p>1.2.5 Recover usable resources from waste.</p> <p>1.2.6 Repair damaged products or materials instead of discarding them.</p> <p>1.3 Use of sustainable materials in RMG production with low environmental impact.</p> <p>1.4 Recycling of materials used in the RMG manufacturing process.</p> <p>1.5 Sustainable transportation methods for moving goods and materials within the RMG sector.</p>                                                                                           |
| 2. Sources of environmental impacts              | <p>2.1 Air Pollution in RMG Manufacturing</p> <p>2.2 Water Pollution in RMG Manufacturing</p> <p>2.3 Soil Degradation in RMG Manufacturing</p> <p>2.4 Noise Pollution in RMG Manufacturing</p> <p>2.5 Waste Generation in RMG Manufacturing</p> <p>2.6 Resource Depletion in RMG Manufacturing</p>                                                                                                                                                                                                                                                                                                                              |
| 3. Ways of mitigating environmental impacts      | <p>3.1 Utilizing energy-efficient equipment in RMG production processes.</p> <p>3.2 Adopting renewable energy sources for RMG manufacturing operations.</p> <p>3.3 Implementing site protection measures to prevent pollution and reduce environmental.</p> <p>3.4 Using reusable materials in production processes.</p> <p>3.5 Choosing sustainable materials with low environmental impact for garment production.</p> <p>3.6 Using noise-reducing equipment and scheduling production work.</p> <p>3.7 Optimizing logistics and delivery to reduce transportation emissions and improve overall supply chain efficiency.</p> |
| 4. Recyclable and non-recyclable items           | <p>4.1 <b>Recyclable Items in RMG Manufacturing</b></p> <p>4.1.1 Metal components.</p> <p>4.1.2 Wooden materials.</p> <p>4.1.3 Fabric scraps and textile fibers.</p> <p>4.1.4 Plastic items, including buttons, tags, and packaging materials.</p> <p>4.2 <b>Non-Recyclable Items in RMG Manufacturing</b></p> <p>4.2.1 Paints, coatings, and dyes.</p> <p>4.2.2 Contaminated materials.</p> <p>4.2.3 Treated wood and other materials.</p>                                                                                                                                                                                     |
| 5. Different types of waste in RMG manufacturing | <p>5.1 Fabric waste from cutting and trimming processes.</p> <p>5.2 Wood waste from packaging or display materials.</p> <p>5.3 Metal waste from trims, buttons, zippers, and other hardware.</p> <p>5.4 Textile waste, including scraps from garment production and defective products.</p>                                                                                                                                                                                                                                                                                                                                     |

|  |                                                                                                                                                                                                                |
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|  | 5.5 Plastic waste, such as packaging materials, labels, and plastic buttons.<br>5.6 Solid waste from chemical treatments or dyeing processes.<br>5.7 Packaging waste from raw materials and finished products. |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Principles of green practices in RMG (Ready-Made Garment) manufacturing.<br>1.2 Sources of environmental impacts.<br>1.3 Ways of mitigating environmental impacts<br>1.4 Recyclable and non-recyclable items<br>1.5 Different types of waste in RMG manufacturing<br>1.6 Hazardous waste in RMG production<br>1.7 Green habits to reduce waste                                               |
| 2. Underpinning Skills    | 2.1 Explaining principles of green practices in RMG manufacturing<br>2.2 Identifying sources of environmental impacts<br>2.3 Explaining re-cyclable and non-recyclable items<br>2.4 Minimizing resources use in the workplace<br>2.5 Identifying different types of waste in RMG manufacturing<br>2.6 Disposing hazardous waste in RMG production<br>2.7 Practicing green habits to reduce waste |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincere and honest to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Communication with peers, subordinates and seniors in the workplace                              |
| 4. Resource Implications  | 4.1 Workplace (simulated or actual)<br>4.2 Projector<br>4.3 Stationery<br>4.4 Learning manual                                                                                                                                                                                                                                                                                                    |

### Assessment Evidence Guide

|                                   |                                                                                                                                     |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 interpreted green concepts<br>1.2 minimizing resources use in the workplace |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|

|                          |                                                                                                                                                                                                                                                                          |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | 1.3 implementing waste management practices                                                                                                                                                                                                                              |
| 2. Methods of Assessment | <p>Competency should be assessed by:</p> <p>2.1 Written test</p> <p>2.2 Practical Demonstration</p> <p>2.3 Oral Questioning</p> <p>2.4 Portfolio (Optional)</p>                                                                                                          |
| 3. Context of Assessment | <p>3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.</p> <p>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</p> |



## The Occupation Specific Competencies

|                                                                                                                                                                                                                                                                                                                               |                                     |                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------|
| <b>Unit of Competency:</b><br><b>INTERPRET ADVANCED PATTERN USING CAD</b>                                                                                                                                                                                                                                                     | <b>Nominal Duration:</b><br>20 Hrs. | <b>Unit Code:</b><br>SICIP-RMG-APM-01-O |
| <b>Unit Descriptor:</b><br><br>This unit covers the skills, knowledge and attitudes required to interpret advanced pattern using CAD. It specifically includes the tasks of understanding advanced pattern making, identifying advanced garment items and identifying critical parts, points and measuring guide of garments. |                                     |                                         |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                                             | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|--------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Understand advanced pattern making                              | 1.1 <b><u>CAD system</u></b> is interpreted as per RMG industry.<br>1.2 Importance of CAD system is described.<br>1.3 <b><u>Functions of CAD</u></b> systems are identified and described.<br>1.4 <b><u>Working sequence</u></b> of advanced CAD section in RMG industry is explained.<br>1.5 <b><u>Strategic benefits</u></b> are described.<br>1.6 <b><u>CAD software</u></b> is identified as per job requirement.                                                                                                                                                                             |
| 2. Identify advanced garment items                                 | 2.1 Advanced garment categories are reviewed based on design complexity and construction techniques.<br>2.2 Garment features are analyzed to determine their functional and aesthetic components.<br>2.3 Style specifications and technical drawings are interpreted from tech packs or CAD files.<br>2.4 Fabric types, trims, and embellishments are identified according to the garment type.<br>2.5 Construction methods and pattern-making techniques are studied for each garment item.<br>2.6 <b><u>Advanced garment items</u></b> are classified and documented as per industry standards. |
| 3. Identify critical parts, points and measuring guide of garments | 3.1 Parts of human body are identified and concisely described.<br>3.2 Job specifications are identified as per requirements.<br>3.3 Measurement process (how to measure) are identified as per product design.<br>3.4 <b><u>Points of measurement</u></b> are identified as per measurement chart.                                                                                                                                                                                                                                                                                               |

### Range of Variables

| Variable | Range (Includes but not limited to): |
|----------|--------------------------------------|
|----------|--------------------------------------|

|                           |                                                                                                                                                                                                                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. CAD system             | 1.1 CAD application and process<br>1.2 Apparel design systems<br>1.3 Fabric prints design / placement<br>1.4 Texture mapping: 3D draping software                                                                                                                                                        |
| 2. Functions of CAD       | 2.1. Pattern making<br>2.2. Grading<br>2.3. Marker making<br>2.4. Consumption<br>2.5. 3D simulation                                                                                                                                                                                                      |
| 3. Working sequence       | 3.1 Collect Technical package/ Specification<br>3.2 Pattern Making<br>3.3 Pattern Grading<br>3.4 Marker making<br>3.5 Marker printing<br>3.6 3D simulation                                                                                                                                               |
| 4. Strategic benefits     | 4.1 Time Efficiency<br>4.2 Cost Reduction<br>4.3 Competitive Advantage<br>4.4 Global Collaboration<br>4.5 Sustainability                                                                                                                                                                                 |
| 5. CAD software           | 5.1 LECTRA<br>5.2 Garber garment technology<br>5.3 Gemini<br>5.4 TUKACAD (Tukatech)<br>5.5 Optitex PDS<br>5.6 GT resources<br>5.7 Winda<br>5.8 Euro CAD                                                                                                                                                  |
| 6. Advanced garment items | <b>Men's wear</b><br>6.1 Casual shirt<br>6.2 Pant / Trouser<br>6.3 Cargo pant<br>6.4 Blazer<br>6.5 Jacket<br><b>Children wear</b><br>6.6 Dress<br>6.7 Jump suite<br>6.8 Romper<br>6.9 Blazer<br>6.10 Jacket<br><b>Lingerie wear</b><br>6.11 Bra<br>6.12 Panty<br>6.13 Nightwear<br>6.14 Swimming costume |
| 7. Points of              | <b>Tops measurement for shirts, jacket, blazers, dresses</b>                                                                                                                                                                                                                                             |

|             |                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| measurement | 7.1 Body length<br>7.2 Chest<br>7.3 Waist<br>7.4 Bottom<br>7.5 Sleeve length<br>7.6 Sleeve opening<br>7.7 Bicep<br>7.8 Shoulder<br>7.9 Armhole<br>7.10 Neck width<br>7.11 Neck depth<br>7.12 Collar width<br>7.13 Collar height<br><b>Bottoms measurement for trouser, pant and skirts</b><br>7.14 Waist<br>7.15 Hip<br>7.16 Thigh<br>7.17 Knee<br>7.18 Leg opening<br>7.19 Front and back rise<br>7.20 Inseam<br>7.21 Out seam |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Computer-aided design tools for creating garment patterns and designs.<br>1.2 Functions include pattern drafting, grading, marking, and simulation.<br>1.3 Order of tasks in garment production, from design to final product.<br>1.4 Benefits include increased efficiency, accuracy, and reduced lead time in production.<br>1.5 Characteristics like fit, fabric type, style, and design elements.<br>1.6 Detailed description of design features, fabric choices, trims, and construction methods.<br>1.7 Complex garments with intricate designs or specialized functions, such as outerwear or activewear.<br>1.8 Key areas for garment fitting, including bust, waist, hips, and inseam.<br>1.9 Guidelines for size, fit, and construction, ensuring consistency and accuracy.<br>1.10 Specific points on the body used to measure garments, such as chest, waist, and length.<br>1.11 A reference chart with standard measurements for various garment sizes. |
| 2. Underpinning Skills    | 2.1 Describing functions of CAD<br>2.2 Listing of working sequence of CAD section                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|                           |                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | 2.3 Listing of garment items<br>2.4 Describing parts of human body<br>2.5 Identifying garment guides<br>2.6 Identifying points of measurement                                                                                                                                                                                                                       |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincere and honest to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Communication with peers, subordinates and seniors in the workplace |
| 4 Resource Implications   | 4.1 Workplace (simulated or actual)<br>4.2 Tools and equipment<br>4.3 Relevant materials<br>4.4 Work instruction<br>4.5 Garments item<br>4.6 Measurement chart                                                                                                                                                                                                      |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                               |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 understood advanced pattern making<br>1.2 identified advanced garment items<br>1.3 identified critical parts, points and measuring guide of garments                                                  |
| 2. Methods of Assessment          | Competency should be assessed by:<br>2.1 Written test<br>2.2 Practical Demonstration<br>2.3 Oral Questioning<br>2.4 Portfolio (Optional)                                                                                                                      |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert. |

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| <b>Unit of Competency:</b><br><b>PERFORM PATTERN MAKING WITH CAD</b>                                                                                                                                                                                                                                                                                                                     | <b>Nominal Duration:</b><br>152 Hrs. | <b>Unit Code:</b><br>SICIP-RMG-APM-02-O |
| <b>Unit Descriptor:</b><br><br>This unit covers the skills, knowledge and attitudes required to perform pattern making with CAD. It specifically includes the tasks of applying basic sewing techniques, performing pattern making for menswear, performing pattern making for children wear, performing pattern making for lingerie wear and maintaining quality issues of the pattern. |                                      |                                         |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                      | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Apply basic sewing techniques            | 1.1 Different <b><u>type of sewing machines</u></b> is identified based on structure and functions.<br>1.2 Various <b><u>stitch types</u></b> are identified by analyzing structure and formation of stitching.<br>1.3 Seams are categorized based on structure, application, and finishing style.<br>1.4 <b><u>Seam types</u></b> are identified based on sample.<br>1.5 Suitability of seams is evaluated for different garment styles and stress points.                                                                                                                                                                                                                                                             |
| 2. Perform pattern making for menswear      | 2.1 Garment specifications and measurements are interpreted from technical sheets or design briefs.<br>2.2 <b><u>Pattern components of menswear</u></b> are selected or drafted according to standard size charts.<br>2.3 Patterns are modified or developed to suit various styles and design features of menswear.<br>2.4 Ease allowance, seam allowance, and fit adjustments are applied based on the garment type.<br>2.5 Pattern components are checked for accuracy and completeness before grading or saving.<br>2.6 Finalized patterns are saved, labeled, and stored in accordance with CAD file management protocols.<br>2.7 Patterns are reviewed and verified against design and construction requirements. |
| 3. Perform pattern making for children wear | 3.1 Garment specifications and measurements are interpreted from technical sheets or design briefs.<br>3.2 <b><u>Pattern components of children wear</u></b> are selected or drafted according to standard size charts.<br>3.3 Patterns are modified or developed to suit various styles and design features of menswear.<br>3.4 Ease allowance, seam allowance, and fit adjustments are applied based on the garment type.<br>3.5 Pattern components are checked for accuracy and completeness before grading or saving.                                                                                                                                                                                               |

|                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                             | <p>3.6 Finalized patterns are saved, labeled, and stored in accordance with CAD file management protocols.</p> <p>3.7 Patterns are reviewed and verified against design and construction requirements.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| 4. Perform pattern making for lingerie wear | <p>4.1 Garment specifications and measurements are interpreted from technical sheets or design briefs.</p> <p>4.2 <b><u>Pattern components of lingerie wear</u></b> are selected or drafted according to standard size charts.</p> <p>4.3 Patterns are modified or developed to suit various styles and design features of menswear.</p> <p>4.4 Ease allowance, seam allowance, and fit adjustments are applied based on the garment type.</p> <p>4.5 Pattern components are checked for accuracy and completeness before grading or saving.</p> <p>4.6 Finalized patterns are saved, labeled, and stored in accordance with CAD file management protocols.</p> <p>4.7 Patterns are reviewed and verified against design and construction requirements.</p> |
| 5. Maintain quality issues of the pattern   | <p>5.1 Pattern accuracy is verified against measurement charts and technical specifications.</p> <p>5.2 Grain lines, notches, and balance marks are checked for correct placement and alignment.</p> <p>5.3 Seam lines, allowances, and ease are reviewed for consistency across all pattern pieces.</p> <p>5.4 Errors in shaping, grading, or symmetry are identified and corrected as per standard guidelines.</p>                                                                                                                                                                                                                                                                                                                                        |

### Range of Variables

| Variable                          | Range (Includes but not limited to):                                                                                                                                            |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Type of sewing machines        | <p>1.1 Single needle lock stitch</p> <p>1.2 Overlock</p> <p>1.3 Flat lock</p>                                                                                                   |
| 2. Stitch types                   | <p>2.1 Chain Stitch</p> <p>2.2 Hand Stitch</p> <p>2.3 Lock Stitch</p> <p>2.4 Multi-thread Chain Stitch</p> <p>2.5 Overlock/Serger Stitch</p> <p>2.6 Cover Stitch (Flatlock)</p> |
| 3. Seam types                     | <p>3.1 Superimposed Seam</p> <p>3.2 Lapped Seam</p> <p>3.3 Bound Seam</p> <p>3.4 Flat Seam</p> <p>3.5 Decorative Seam</p> <p>3.6 Edge Finishing Seam</p>                        |
| 4. Pattern components of menswear | <p>4.1 T-shirt</p> <p>4.2 Polo shirt</p> <p>4.3 Shirts</p>                                                                                                                      |

|                                        |                                           |
|----------------------------------------|-------------------------------------------|
|                                        | 4.4 Trouser<br>4.5 Jogger pant            |
| 5. Pattern components of children wear | 5.1 Dress<br>5.2 Jump suite<br>5.3 Romper |
| 6. Pattern components of lingerie wear | 6.1 Bra<br>6.2 Panty                      |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | 1.1 Machines like lockstitch, overlock, and cover stitch, each for specific tasks.<br>1.2 Types include lockstitch, chainstitch, zigzag, and blind stitch.<br>1.3 Types like plain, double-stitched, French, and flat-felled seams.<br>1.4 Detailed features like size, fit, design elements, and fabric choice.<br>1.5 Software used to create and modify garment patterns for production (Pattern design system (PDS)).<br>1.6 A reference chart outlining various sizes and corresponding style measurements.<br>1.7 Parts of men's wear.<br>1.8 Parts of children wear.<br>1.9 Parts of lingerie wear.<br>1.10 Problems like incorrect measurements, misalignment, or fabric waste during pattern creation. |
| 2. Underpinning Skills    | 2.1 Following size tables and styles<br>2.2 Creating parts of men's wear<br>2.3 Calculating sewing allowance and shrinkage<br>2.4 Performing variant and exporting process<br>2.5 Converting pattern making systems<br>2.6 Maintaining quality issues of pattern                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincere and honest to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Communication with peers, subordinates and seniors in the workplace                                                                                                                                                                                                                                                                                                                                             |
| 4 Resource Implications   | 4.1 Personal Protective Equipment (PPE)<br>4.2 Tools and equipment<br>4.3 Paper<br>4.4 Pen                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                                                                                                                                       |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | <p>Assessment required evidence that the candidate:</p> <ul style="list-style-type: none"> <li>1.1 applied basic sewing techniques</li> <li>1.2 performed pattern making for menswear</li> <li>1.3 performed pattern making for children wear</li> <li>1.4 performed pattern making of lingerie wear</li> <li>1.5 maintained quality issues of the pattern</li> </ul> |
| 2. Methods of Assessment          | <p>Competency should be assessed by:</p> <ul style="list-style-type: none"> <li>2.1 Written test</li> <li>2.2 Practical Demonstration</li> <li>2.3 Oral Questioning</li> <li>2.4 Portfolio (Optional)</li> </ul>                                                                                                                                                      |
| 3. Context of Assessment          | <ul style="list-style-type: none"> <li>3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.</li> <li>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.</li> </ul>                                                 |



|                                                                                                                                                                                                                                                                                 |                                     |                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|-----------------------------------------|
| <b>Unit of Competency:</b><br><b>PERFORM PATTERN GRADING</b>                                                                                                                                                                                                                    | <b>Nominal Duration:</b><br>20 Hrs. | <b>Unit Code:</b><br>SICIP-RMG-APM-03-O |
| <b>Unit Descriptor:</b><br><br>This unit covers the skills, knowledge and attitudes required to perform pattern grading. It specifically includes the tasks of interpreting size and measurement differences, applying rule table for grading and carrying out pattern grading. |                                     |                                         |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                        | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Interpret size and measurement differences | 1.1 Measurement charts from various sources are reviewed and explained.<br>1.2 Body measurements and finished garment measurements are analyzed.<br>1.3 Differences between international, regional, and brand-specific size standards are identified.<br>1.4 Adjustments to pattern measurements are made based on sizing requirements.<br>1.5 Fit issues arising from measurement discrepancies are identified and interpreted.<br>1.6 Size specifications are documented accurately for use in pattern development and grading.                                                                                               |
| 2. Apply rule table for grading               | 2.1 Grading rules are selected according to garment specifications.<br>2.2 Size increments and <b><u>grading points</u></b> are identified and applied to the base pattern using CAD tools.<br>2.3 Rule table values are verified to ensure consistent grading across all pattern pieces.<br>2.4 Graded patterns are checked for accuracy in alignment, fit, and balance.<br>2.5 Errors in grade application are corrected based on inspection or feedback.<br>2.6 Grading data is saved and documented as per standard operating procedures.<br>2.7 Rule tables are updated or customized when non-standard sizing is required. |
| 3. Carry out pattern grading                  | 3.1 Graded patterns are collected for marker making<br>3.2 Graded patterns are prepared as per measurement chart<br>3.3 Prepared patterns are saved and stored<br>3.4 Prepared patterns are cut using <b><u>Pattern Cutter Machine</u></b>                                                                                                                                                                                                                                                                                                                                                                                       |

## Range of Variables

| Variable                  | Range (Includes but not limited to):                                                     |
|---------------------------|------------------------------------------------------------------------------------------|
| 1. Grading points         | 1.1 Chest<br>1.2 Length<br>1.3 Sleeve<br>1.4 Arm hole<br>1.5 Neck width<br>1.6 Neck drop |
| 2. Pattern Cutter Machine | 2.1. Bed system<br>2.2. Plotter system                                                   |

## Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| 1. Underpinning Knowledge | 1.1 Sizes represented on patterns, reflecting body measurements and garment fit.<br>1.2 A chart outlining size grading rules, including measurements for each size increment.<br>1.3 The process of adjusting pattern dimensions to create different sizes.<br>1.4 Patterns modified according to specific size increments, ready for production.<br>1.5 The axis (horizontal/vertical lines) guides the even scaling of pattern pieces during grading. |
| 2. Underpinning Skills    | 2.1 Calculating measurement and sizes<br>2.2 Applying appropriate rule table for grading<br>2.3 Checking graded pattern<br>2.4 Collecting graded patterns<br>2.5 Preparing graded patterns                                                                                                                                                                                                                                                              |
| 3. Underpinning Attitudes | 3.1. Commitment to occupational health and safety<br>3.2. Promptness in carrying out activities<br>3.3. Sincere and honest to duties<br>3.4. Environmental concerns<br>3.5. Eagerness to learn<br>3.6. Tidiness and timeliness<br>3.7. Respect for rights of peers and seniors in the workplace<br>3.8. Communication with peers, subordinates and seniors in the workplace                                                                             |
| 3 Resource Implications   | 3.1 Workplace (simulated or actual)<br>3.2 Manuals<br>3.3 Graded patterns<br>3.4 Paper<br>3.5 Pen                                                                                                                                                                                                                                                                                                                                                       |

## Assessment Evidence Guide

|                                   |                                                                                                      |
|-----------------------------------|------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 interpreted size and measurement differences |
|-----------------------------------|------------------------------------------------------------------------------------------------------|

|                          |                                                                                                                                                                                                                                                               |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                          | 1.2 applied rule table for grading<br>1.3 carried out pattern grading                                                                                                                                                                                         |
| 2. Methods of Assessment | Competency should be assessed by:<br><br>2.1 Written test<br>2.2 Practical Demonstration<br>2.3 Oral Questioning<br>2.4 Portfolio (Optional)                                                                                                                  |
| 3. Context of Assessment | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert. |

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| <b>Unit of Competency:</b><br><b>MAKE MARKER CONSUMPTION AND PRINTING</b>                                                                                                                                                                                             | <b>Nominal Duration:</b><br>30 Hrs. | <b>Unit Code:</b><br>SICIP-RMG-APM-04-O |
| <b>Unit Descriptor:</b><br><br>This unit covers the skills, knowledge and attitudes required to make marker consumption and printing. It specifically includes the tasks of making marker, calculating consumption and printing and plotting marker using CAD system. |                                     |                                         |

### Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

| Elements of Competency                    | Performance Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Make marker                            | 1.1 Pattern pieces are imported or selected from the database using the CAD system.<br>1.2 Garment styles and size ranges are confirmed before marker creation.<br>1.3 Fabric width and type are set according to production requirements.<br>1.4 Pattern pieces are arranged efficiently to optimize fabric utilization and minimize wastage.<br>1.5 Marker is created according to standard layout rules and cutting specifications.<br>1.6 Marker layout is saved, named, and stored as per organizational protocols.<br>1.7 <b><u>Generalities of marker and composition table</u></b> are filled with necessary information<br>1.8 Final marker is reviewed and approved before printing or plotting. |
| 2. Calculate consumption                  | 2.1 <b><u>Fabric types</u></b> and widths are identified and selected based on the given pattern and design requirements.<br>2.2 Garment components are analyzed and their consumption is estimated according to standard practices.<br>2.3 Patterns are arranged efficiently on marker layouts to minimize fabric wastage.<br>2.4 Seam allowances, shrinkage percentages, and fabric defects are considered while calculating consumption.<br>2.5 Fabric consumption is computed using CAD software tools as per industry standards.<br>2.6 Final consumption data is reviewed and verified before marker plotting.                                                                                       |
| 3. Print and plot marker using CAD system | 3.1 Marker files are opened and verified using the CAD system.<br>3.2 Plotter settings are configured according to marker dimensions and fabric width.<br>3.3 Print/plot commands are executed as per the specified                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |

|  |                                                                                                                                                                                                                                                                                                                                    |
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|  | <p>layout and paper type.</p> <p>3.4 Markers are printed or plotted without distortion and within defined tolerances.</p> <p>3.5 Output is monitored to ensure accuracy and continuity during the plotting process.</p> <p>3.6 Printed/plotted markers are labeled, stored, or dispatched following organizational procedures.</p> |
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### Range of Variables

| Variable                                        | Range (Includes but not limited to):                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Generalities of marker and composition table | <p>1.1 Name</p> <p>1.2 Comments</p> <p>1.3 Size</p> <p>1.4 Quantity</p> <p>1.5 Width</p> <p>1.6 Length</p> <p>1.7 Salvage value</p> <p>1.8 Marker type</p> <p>1.8.1 Solid marker</p> <p>1.8.2 Salvage marker</p> <p>1.8.3 Single direction marker</p> <p>1.8.4 All garments one way</p> <p>1.8.5 One garments one way</p> <p>1.8.6 Grouped marker</p> <p>1.9 Efficiency</p> <p>1.10 Fabric edges</p> <p>1.11 Moving tolerance</p> <p>1.12 Fine rotation</p> <p>1.13 Model name</p> <p>1.14 Save</p> |
| 2. Marker making                                | <p>2.1 Pop and drag</p> <p>2.2 Auto nesting</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 3. Fabric types                                 | <p>3.1 Knit</p> <p>3.2 Woven</p> <p>3.3 Denim</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

### Curricular Content Guide

|                           |                                                                                                                                                                                                                                                                                                                                                                                                       |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Underpinning Knowledge | <p>1.1 Software used to create efficient layouts for fabric cutting, minimizing waste.</p> <p>1.2 Software used to control printers for transferring marker layouts onto fabric.</p> <p>1.3 Markers include pattern pieces arranged optimally on fabric; must account for fabric width and grain.</p> <p>1.4 The process of placing pattern pieces on a fabric layout to optimize material usage.</p> |
|---------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

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|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | 1.5 Steps to print the marker on fabric, including alignment, transfer techniques, and quality checks.                                                                                                                                                                                                                                                              |
| 2. Underpinning Skills    | 2.1 Collecting marker<br>2.2 Preparing marker<br>2.3 Installing printer software<br>2.4 Preparing printer<br>2.5 Performing marker making<br>2.6 Performing marker print                                                                                                                                                                                            |
| 3. Underpinning Attitudes | 3.1 Commitment to occupational health and safety<br>3.2 Promptness in carrying out activities<br>3.3 Sincere and honest to duties<br>3.4 Environmental concerns<br>3.5 Eagerness to learn<br>3.6 Tidiness and timeliness<br>3.7 Respect for rights of peers and seniors in the workplace<br>3.8 Communication with peers, subordinates and seniors in the workplace |
| 4 Resource Implications   | 4.1 Workplace (simulated or actual)<br>4.2 Manuals<br>4.3 Printer<br>4.4 Marker<br>4.5 Paper<br>4.6 Pen                                                                                                                                                                                                                                                             |

### Assessment Evidence Guide

|                                   |                                                                                                                                                                                                                                                               |
|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Critical Aspects of Competency | Assessment required evidence that the candidate:<br>1.1 made marker<br>1.2 calculated consumption<br>1.3 printed and plotted marker using CAD system                                                                                                          |
| 2. Methods of Assessment          | Competency should be assessed by:<br>2.1 Written test<br>2.2 Practical Demonstration<br>2.3 Oral Questioning<br>2.4 Portfolio (Optional)                                                                                                                      |
| 3. Context of Assessment          | 3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training.<br>3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert. |

### End of the Competency Standard

### Workshop/Lab Facility Standard

|                            |                                  |
|----------------------------|----------------------------------|
| <b>Course Name:</b>        | Advanced Pattern Making with CAD |
| <b>Number of Trainees:</b> | 20                               |

**Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):**

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)
- OR
- Classroom cum workshop – 1000 sft (93 sqm)

**Major Training Equipment and Training Facilities:**

| S. N. | Major Equipment and Training facilities                                                 | Required facilities |
|-------|-----------------------------------------------------------------------------------------|---------------------|
| 1.    | Computers/Laptop (Minimum Core i3, 8GB RAM, 500GB SSD, 21" Monitor, Graphics-supported) | 20                  |
| 2.    | CAD Software (Lectra/Gerber/Optitex/Gemini)                                             | 1                   |
| 3.    | UPS/Backup Power Supply (1–2 hours backup per workstation)                              | 20                  |
| 4.    | Plotter/Printer (A0 size, compatible with CAD software, minimum 40-60 inches width)     | 1                   |
| 5.    | Projector & Screen (HD quality, HDMI support, 3000+ lumens)                             | 1                   |
| 6.    | Mannequins/Dress Forms (Female and male standard sizes)                                 | 2                   |
| 7.    | Working Table (Minimum size 2 ft × 4 ft, laminated surface)                             | 2                   |
| 8.    | Single needle lock stitch machine                                                       | 1                   |
| 9.    | Overlock machine                                                                        | 1                   |
| 10.   | Flat lock machine                                                                       | 1                   |

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Advanced Pattern Making with CAD, the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the

remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 20 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 20 trainees.